ALGEBRA 1 · UNIT 4: RELATIONS AND FUNCTIONS

# Roller Coaster Designer

*Use functions to analyze a coaster ride and design your own.*

**Honors**

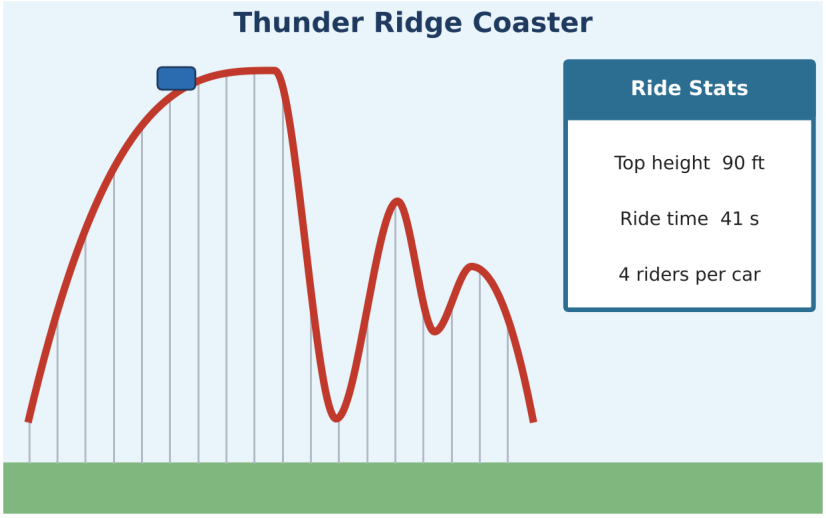
2–3 class periods

Name: _____ Date: _____ Period: _____

## The Scenario

Thunder Ridge Park is building a new roller coaster, and your team has been hired as ride designers. Before you build your own coaster, you need to study the park's current ride, Thunder Ridge.

Engineers describe a coaster with functions: the height of a car depends on time, the wait in line depends on your place, and every rider is assigned to exactly one car. You will analyze these functions and then design a coaster that meets the park's rules.




## Learning Goals

- I can represent a graph as a piecewise function and solve equations on each piece.
- I can connect key features and rates of change to context.
- I can design a function that satisfies several constraints.

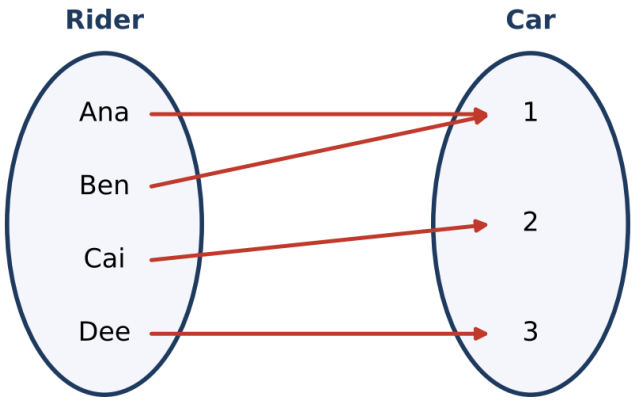
## Project Timeline

| When | What you will do |
|---|---|
| Day 1 | Functions & Relations |
| Day 2 | The Piecewise Ride |
| Day 3 | Design Within the Rules |
| Final | Coaster Design Proposal + reflection |

## Part 1: Functions & Relations

**Day 1 ·** Justify each answer.

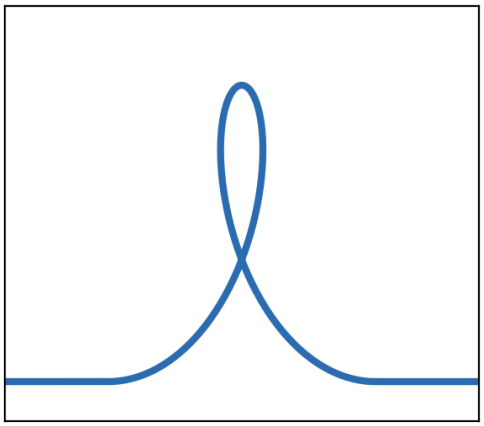
**1.** Explain why "rider → car" is a function but its inverse relation is not. What would have to be true for the inverse to be a function?



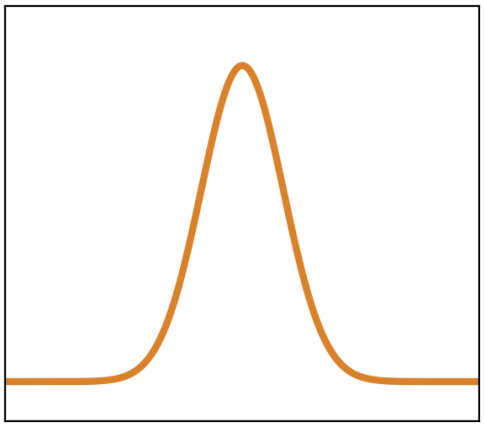

**2.** For which values of k is {(1, 4), (2, k), (k, 8)} a function?

**3.** Explain, with the vertical line test, why the loop track is not a function of horizontal position, and describe a different input that would make the loop a function.

Track A: height vs. horizontal position



Track B: height vs. time

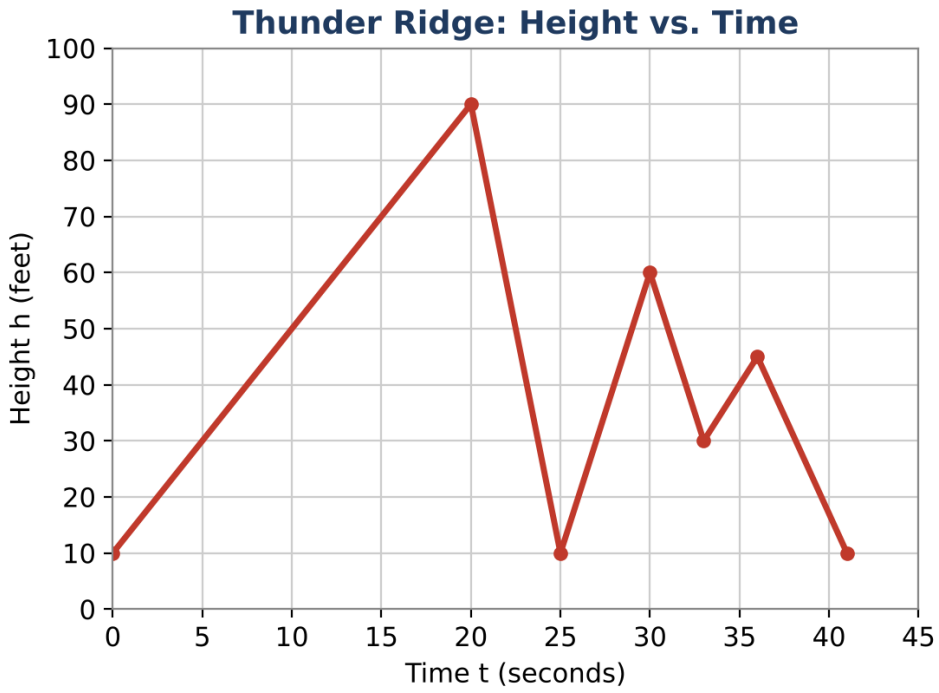


**4.** Wristband: 8 rides for $20, or $3 per ride without a band. Write a rule for the cheapest cost C(r) for r rides ($0 \le r \le 8$) and give its range.

## Part 2: The Piecewise Ride

**Day 2 ·** Use the Thunder Ridge graph.

**5.** Write h(t) as a piecewise function with six linear pieces (include each domain).




**6.** Find every time t when h(t) = 45. Show which pieces you used.

**7.** Find the average rate of change on each piece. Which piece is the fastest climb? The fastest drop?

**8.** Park rule: no drop may be faster than 15 ft per second on average. Does Thunder Ridge follow it? If not, propose a new point that fixes it without changing the top height.

## Part 3: Design Within the Rules

**Day 3 ·** Rules: 30 to 60 s; top height 100 to 150 ft; at least three hills; no climb faster than 6 ft per s; no drop faster than 15 ft per s; start and end at 10 ft.

**9.** Design your coaster as a piecewise linear function. List every point and every piece.

**10.** Show that every piece follows the rate rules. Give the domain, range, and all extrema.

**11.** The queue for your ride grows by 1.5 minutes per group starting at 2 minutes, and your ride loads a new group every 50 seconds. Write the wait as a function of group number and decide whether the 1.5-minute growth is realistic.

## Final Product: Coaster Design Proposal

**Your final product must include:**

- Piecewise function for Thunder Ridge
- All solutions to h(t) = 45
- Safety analysis and fix
- Your piecewise coaster with checked rates
- Queue model and reflection

## Reflection

**R1.** How does a piecewise function let you describe a complicated graph exactly?

**R2.** How did the rate rules limit your design choices?

## Scoring Rubric

| Criteria | 4 – Exceeds | 3 – Meets | 2 – Approaching | 1 – Beginning |
|---|---|---|---|---|
| **Mathematical Accuracy** | All calculations and solutions are correct; answers include units and are checked. | Nearly all work is correct; minor errors do not change conclusions. | Several errors; some conclusions are affected. | Many errors or missing work. |
| **Reasoning & Justification** | Clear, complete explanations connect every answer to the situation; uses precise vocabulary. | Explanations are mostly clear and connected to the situation. | Explanations are brief or partly unclear. | Little or no explanation. |
| **Application & Design** | Coaster design meets every park rule and its key features are identified correctly. Goes beyond the minimum with a creative, realistic choice. | Coaster design meets every park rule and its key features are identified correctly. | Design is incomplete or only partly uses the math from the unit. | Design is missing or unrelated to the unit math. |
| **Communication & Presentation** | Proposal includes a labeled graph, domain and range, key features, function notation, and the queue model. Polished, organized, and visually engaging. | Proposal includes a labeled graph, domain and range, key features, function notation, and the queue model. Organized and easy to follow. | Product is missing required parts or is hard to follow. | Product is mostly missing. |